Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

When a ceiling fan is switched off, its angular velocity falls to half while it makes 36 rotations. How many more rotations will it make before coming to rest?

(A) 24

(B) 36

(C) 18

(D) 12

Q.2

The moment of inertia of a body about a given axis is $1.2\ \mathrm{kg\ m^2}$. Initially, the body is at rest. In order to produce a rotational kinetic energy of 1500 joule, an angular acceleration of 25 radian $/\mathrm{sec}^2$ must be applied about that axis for a duration of

(A) 4 seconds

(B) 2 seconds

(C) 8 seconds

(D) 10 seconds

Q.3

A pulley of radius $2\ \mathrm{m}$ is rotated about its axis by a force $F = \left(20t - 5t^2\right)$ newton (where t is measured in seconds) applied tangentially. If the moment of inertia of the pulley about its axis of rotation is $10\ \mathrm{kg} - \mathrm{m}^2$ the number of rotations made by the pulley before its direction of motion is reversed, is:

(A) more than 3 but less than 6

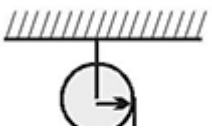
(B) more than 6 but less than 9

(C) more than 9

(D) less than 3

Q.4

Two identical discs of mass m and radius r are arranged as shown in the figure. If α is the $\underline{\lim}$ angular acceleration of the lower disc and $\mathrm{a_{cm}}$ is acceleration of centre of mass of the lower disc, then relation between $\mathrm{a_{cm}}$, α and r is



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(A) $a_{cm} = \alpha/r$

(B) $a_{cm} = 2\alpha r$

(C) $a_{cm} = \alpha r$

(D) None of these

Q.5

A loop of radius r and mass m rotating with an angular velocity ω_0 is placed on a rough horizontal surface. The initial velocity of the centre of the hoop is zero. What will be the velocity of the centre of the hoop when it ceases to slip?

(A) $\frac{r\omega_0}{4}$

(B) $\frac{r\omega_0}{3}$

(C) $\frac{r\omega_0}{2}$

(D) $r\omega_0$

Q.6

A ball rolls without slipping. The radius of gyration of the ball about an axis passing through its centre of mass is K. If radius of the ball be R, then the fraction of total energy associated with its rotational energy will be

(A) $\frac{K^2}{R^2}$

(B) $\frac{K^2}{K^2+R^2}$

(C) $\frac{R^2}{K^2+R^2}$

(D) $\frac{K^2+R^2}{R^2}$

Q.7

A ring of mass M and radius R is rotating about its axis with angular velocity ω. Two identical bodies each of mass m are now gently attached at the two ends of a diameter of the ring. Because of this, the kinetic energy loss will be :

(A) $\frac{m(M+2m)}{M}\omega^2R^2$

(B) $\frac{Mm}{(M+m)}\omega^2R^2$

(C) $\frac{Mm}{(M+2m)}\omega^2R^2$

(D) $\frac{(M+m)M}{(M+2m)}\omega^2R^2$

Q.8

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When a ceiling fan is switched off, its angular velocity falls to half while it makes 36 rotations. How many more rotations will it make before coming to rest?

(A) 24

(B) 36

(C) 18

(D) 12

Q.9

A rod PQ of length Lrevolves in a horizontal plane about the axis YY'. The angular velocity of the rod is ω. If A is the area of cross-section of the rod and ρ be its density, its rotational kinetic energy is

(A) $\frac{1}{3}\mathrm{AL}^3\rho\omega^2$

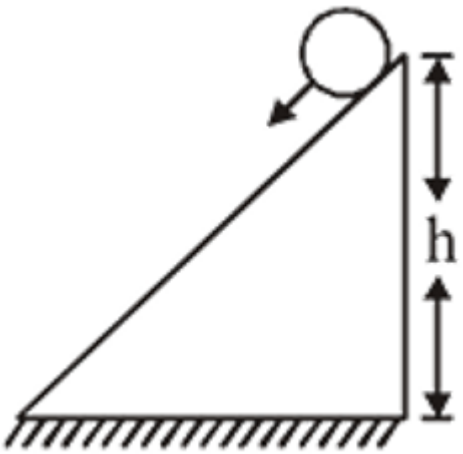
(B) $\frac{1}{2}\mathrm{AL}^3\rho\omega^2$

(C) $\frac{1}{24}\mathrm{AL}^3\rho\omega^2$

(D) $\frac{1}{18}\mathrm{AL}^3\rho\omega^2$

Q.10

A solid cylinder of mass $\mathrm{m\&}$ radius R rolls down inclined plane without slipping. The speed of its C.M. when it reaches the bottom is



(A) $\sqrt{2\mathrm{gh}}$

(B) $\sqrt{4\mathrm{gh}/3}$

(C) $\sqrt{3/4\mathrm{gh}}$

(D) $\sqrt{4\mathrm{gh}}$

Q.11

The least coefficient of friction for an inclined plane inclined at angle α with horizontal in order that a solid cylinder will roll down without slipping is

(A) $\frac{2}{3}\tan\alpha$

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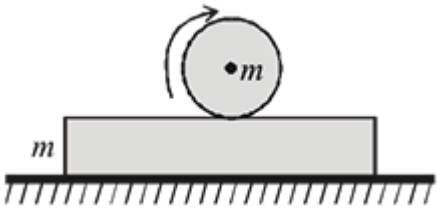
(B) $\frac{2}{7}\tan\alpha$

(C) $\tan\alpha$

(D) $\frac{5}{7}\tan\alpha$

Q.12

A sphere of mass ' m ' is given some angular velocity about a m horizontal axis through its π centre and gently placed on a plank of mass ' m '. The coefficient of friction between the two is μ. The plank rests on a smooth horizontal surface. The initial acceleration of the plank is



(A) zero

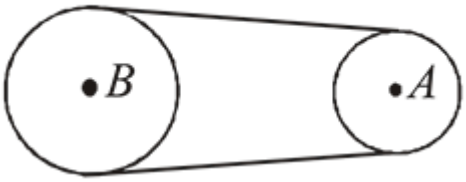
(B) (7 / 5) μg

(C) μg

(D) $2\mu g$

Q.13

A wheel of radius $0.1\ \mathrm{m}$(wheel A) is attached by non-stretching belt to a wheel of radius $0.2\ \mathrm{m}$ (wheel B). The belt does not slip. By the time wheel B turns through 1 revolution, wheel A will rotate through



(A) $\frac{1}{2}$

(B) 1 revolution

(C) 2 revolution

(D) 4 revolution

Q.14

A solid sphere slides down a smooth inclined plane of inclination θ with acceleration a_1. A disc rolls down a rough inclined plane of inclination θ with acceleration a_2. Value of $\frac{a_1}{a_2}$ is

(A) $\frac{3}{2}$

(B) $\frac{2}{3}$

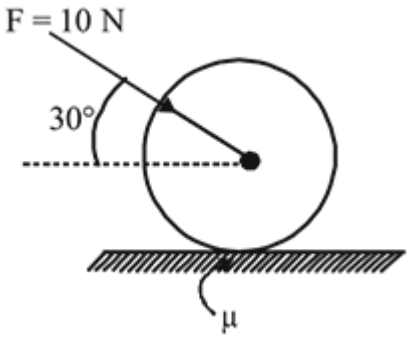
(C) $\frac{1}{2}$

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(D) 2

Q.15

A hollow sphere of mass $2\ \mathrm{F\ kg}$ is kept on a rough horizontal surface. A force of $10\ \mathrm{N}$ is applied at the centre of the sphere as shown in the figure. Find
the minimum value of μ so that the sphere starts pure rolling. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)



(A) $\sqrt{3} \times 0.16$

(B) $\sqrt{3} \times 0.08$

(C) $\sqrt{3} \times 0.1$

(D) Data insufficient

Q.16

A wheel having angular momentum $2\pi \mathrm{kg} - \mathrm{m^2/s}$ about its vertical axis, rotates at the rate of $60\mathrm{rpm}$ about this axis, The torque which can stop the wheel's rotation in 30 sec would be

(A) $\frac{2\pi}{15}\mathrm{Nm}$

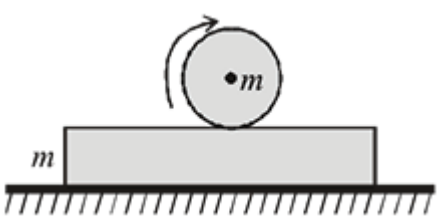
(B) $\frac{\pi}{18}\mathrm{Nm}$

(C) $\frac{\pi}{12}\mathrm{Nm}$

(D) $\frac{\pi}{15}\mathrm{Nm}$

Q.17

A sphere of mass ' m ' is given some angular velocity about a m horizontal axis through its π centre and gently placed on a plank of mass ' m '. The coefficient of friction between the two is μ. The plank rests on a smooth horizontal surface. The initial acceleration of the plank is



(A) zero

(B) (7 / 5) $\mu \mathrm{g}$

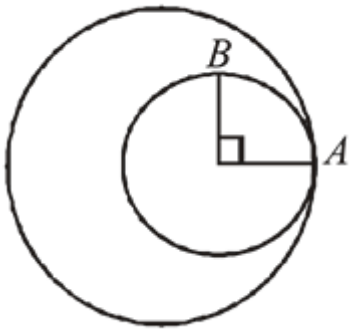
(C) $\mu \mathrm{g}$

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(D) $2\mu g$

Q.18

A and B are moving in circular orbits with angular velocity 2ω and ω respectively. Their positions are as shown at $t = 0$. Find the time when they will meet for the first time.



(A) $\frac{\pi}{2\omega}$

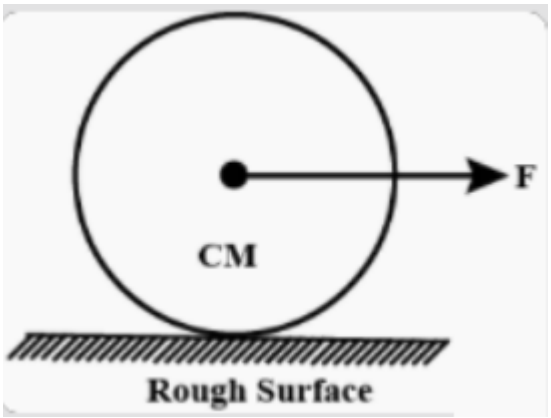
(B) $\frac{3\pi}{2\omega}$

(C) $\frac{\pi}{\omega}$

(D) they will never meet

Q.19

A solid sphere of mass M and radius R is pulled horizontally on a rough surface as shown as shown in figure. Choose the correct alternatives.



(A) The magnitude of the frictional force is $\mathrm{F}/3$

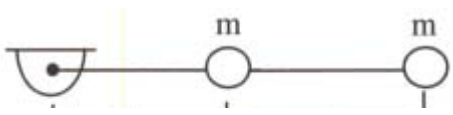
(B) The frictional force on the sphere acts forward

(C) The acceleration of the centre of mass is F/M

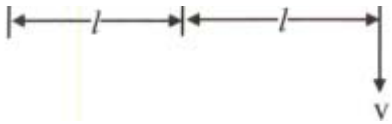
(D) none of these

Q.20

Two beads each of mass m are fixed at the ends of two light rigid rods each of length l as shown in the figure. If the pivot is smooth, Calculate the ratio of translational and rotational kinetic energy of system.



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(A) 8

(B) 7

(C) 6

(D) 9

Answers & Solutions

Q.1 Answer:

12

Solution:

No solution available

Q.2 Answer:

2 seconds

Solution:

$$\begin{aligned}
&(\mathrm{b})\ \mathrm{I}=1.2\ \mathrm{kg\ m^2}, \mathrm{E_r}=1500\ \mathrm{J}\\
&\alpha=25\mathrm{rad/sec^2}, \omega_1=0, \mathrm{t}=?\\
&\text{As } \mathrm{E_r}=\tfrac{1}{2}\mathrm{I}\omega^2\\
&\omega=\sqrt{\tfrac{2\mathrm{E_r}}{\mathrm{I}}}=\sqrt{\tfrac{2\times1500}{1.2}}=50\mathrm{rad/sec}\\
&\text{From } \omega_2=\omega_1+\alpha t\\
&50=0+25\mathrm{t}, \quad \therefore \quad \mathrm{t}=2\ \text{seconds}
\end{aligned}$$

Q.3 Answer:

more than 3 but less than 6

Solution:

$$\begin{aligned}
&F=20t-5t^2\\
&\therefore \alpha=\tfrac{FR}{I}=4t-t^2\\
&\Rightarrow \tfrac{d\omega}{dt}=4t-t^2\\
&\Rightarrow \int_0^{\omega} d\omega=\int_0^t\left(4t-t^2\right)dt\\
&\Rightarrow \omega=2t^2-\tfrac{t^3}{3}(\text{ as } \omega=0 \text{ at } t=0,6\ \mathrm{s})\\
&\int_0^{\theta} d\theta=\int_0^6\left(2t^2-\tfrac{t^3}{3}\right)dt\\
&\Rightarrow \theta=36\mathrm{rad}\\
&\Rightarrow n=\tfrac{36}{2\pi}<6
\end{aligned}$$

Q.4 Answer:

$a_{\mathrm{cm}}=2\alpha \mathrm{r}$

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Solution:

$\mathrm{Tr} = \frac{\mathrm{mr}^2}{2}\alpha_1$

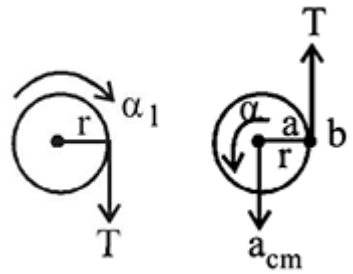
. . .. . .

(1)

$\mathrm{Tr} = \frac{mr^2}{2}\alpha \quad \ldots\ldots$

$\alpha_1 = \alpha$

. . .. . .



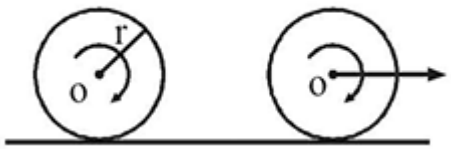
Acceleration of point b **= acceleration of point a .....** ($\mathrm{r}\alpha_1 = \mathrm{a_{cm}} - \mathrm{r}\alpha$

(4) Hence, $2\mathrm{r}\alpha = \mathrm{a_{cm}}$

Q.5 Answer:

$\frac{\mathrm{r}\omega_0}{2}$

Solution:



From conservation of angular momentum about any fix point on the surface, $mr^2\omega_0 = 2mr^2\omega$

$\Rightarrow \omega = \omega_0/2 \Rightarrow v = \frac{\omega_0 r}{2}[\because v = r\omega]$

Q.6 Answer:

$\frac{\mathrm{K}^2}{\mathrm{K}^2+\mathrm{R}^2}$

Solution:

$$\frac{\text{Rotational KE}}{\text{Total KE}} = \frac{\frac{1}{2}\mathrm{mv}^2\left(\frac{\mathrm{K}^2}{\mathrm{R}^2}\right)}{\frac{1}{2}\mathrm{mv}^2\left(1+\frac{\mathrm{K}^2}{\mathrm{R}^2}\right)} = \frac{\mathrm{K}^2}{\mathrm{K}^2+\mathrm{R}^2}$$

Q.7 Answer:

$\frac{Mm}{(M+2m)}\omega^2 R^2$

Solution:

Kinetic energy (rotational) $\mathrm{K_R} = \frac{1}{2}\mathrm{I}\omega^2$

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Kinetic energy (translational) $\mathrm{K_T} = \frac{1}{2}\mathrm{Mv}^2$
$(\mathrm{v} = \mathrm{R}\omega)$
M.I. (initial) $I_{\text{ring}} = MR^2; \omega_{\text{initial}} = \omega$ **M.I.** (new) $\mathrm{I'_{(system)}} = \mathrm{MR}^2 + 2\mathrm{mR}^2$
$\omega'_{(\text{system})} = \frac{\mathrm{M}\omega}{\mathrm{M}+2\,\mathrm{m}}$
Solving we get loss in K.E. $= \frac{\mathrm{Mm}}{(\mathrm{M}+2\,\mathrm{m})}\omega^2\mathrm{R}^2$

Q.8 Answer:

12

Solution:

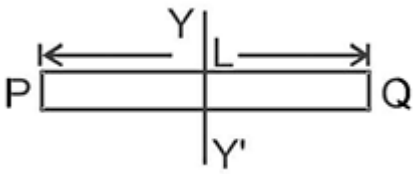
No solution available

Q.9 Answer:

$\frac{1}{24}\mathrm{AL}^3\rho\omega^2$

Solution:

If rotation axis is passing through its middle point & is $\perp$ to its plane, then moment of inertia about YY' is



$I = \frac{ML^2}{12}$ **where** $M =$ **volume** $\times$ **density** $= (L \times A) \times \rho$
so $I = \frac{L^3 A\rho}{12}$
so rotational $\mathrm{K.E} = \frac{1}{2}\mathrm{I}\omega^2 = \frac{\mathrm{L}^3\,\mathrm{A}\rho\omega^2}{24}$

Q.10 Answer:

$\sqrt{4\mathrm{gh}/3}$

Solution:

By energy conservation $(\mathrm{K.E})_\mathrm{i} + (\mathrm{P.E})_\mathrm{i} = (\mathrm{K}\cdot\mathrm{E})_\mathrm{f} + (\mathrm{P}\cdot\mathrm{E})_\mathrm{f}$
$(\mathrm{K.E})_\mathrm{i} = 0, (\mathrm{P.E})_\mathrm{i} = \mathrm{mgh}, (\mathrm{P.E})_\mathrm{f} = 0$
$(\mathrm{K.E})_\mathrm{f} = 1/2\mathrm{I}\omega^2 + 1/2\mathrm{mv}_{\mathrm{cm}}^2$
Where I (moment of inertia) $= 1/2\mathrm{mR}^2$ **(for solid cylinder) so** $mgh = 1/2\left(1/2mR^2\right)\left(\frac{v_{\mathrm{cm}}^2}{R^2}\right) + 1/2mv_{\mathrm{cm}}^2$
$\Rightarrow v_{\mathrm{cm}} = \sqrt{4\mathrm{gh}/3}$

Q.11 Answer:

$\tan\alpha$

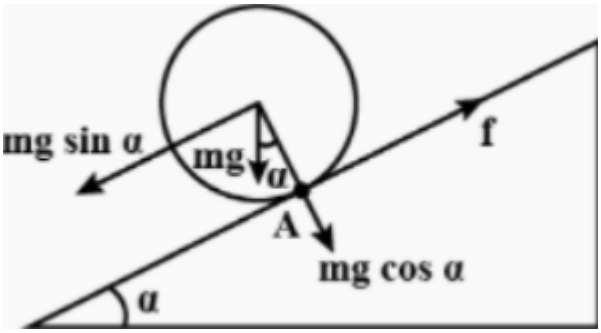
Solution:

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Acceleration of point A must be zero due to pure rolling. Thus $mg\sin\alpha - f = 0$ **for point** A

$\text{mgsin}\,\alpha - \mu\text{mg}\cos\alpha = 0$

$\implies \mu = \tan\alpha$



Q.12 Answer:

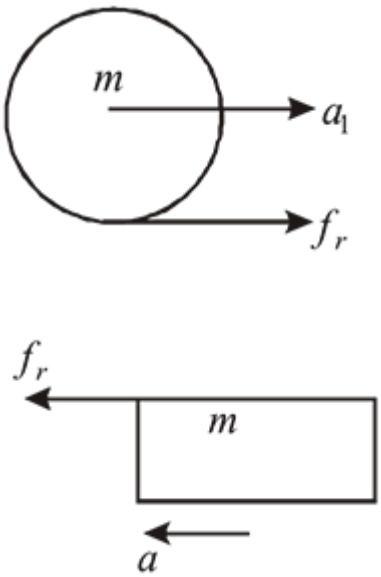
μg

Solution:

FBD for sphere and block $a_1 = \frac{f_r}{m} = \frac{\mu mg}{m}, a_2 = \frac{f_r}{m} = \frac{\mu mg}{m}$

$\vec{a}_1 = \mu g\hat{i}, \vec{a}_2 = -\mu g\hat{i}$

$\overset{f_r}{\longleftarrow} a$



Q.13 Answer:

2 revolution

Solution:

Let speed of belt be v **Angular speeds of wheels** $\omega_B = \frac{v}{2\pi R_B}, \omega_A = \frac{v}{2\pi R_A}; \frac{\omega_A}{\omega_B} = \frac{R_B}{R_A} = 2$

Q.14 Answer:

$\frac{3}{2}$

Solution:

Acceleration of solid sphere sliding down an inclined plane, $a_1 = g\sin\theta$

For disc plane is rough, so it will roll down the incline and its acceleration in rolling is given as, $a_2 =$

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$$\frac{g\sin\theta}{1+\frac{I}{MR^2}} = \frac{g\sin\theta}{1+\frac{\frac{MR^2}{2}}{MR^2}} = \frac{2g\sin\theta}{3}$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{g\sin\theta}{\frac{2g\sin\theta}{3}} = \frac{3}{2}$$

Q.15 Answer:

$\sqrt{3} \times 0.08$

Solution:

$10\cos 30^\circ - f = 2a$... **(i)**

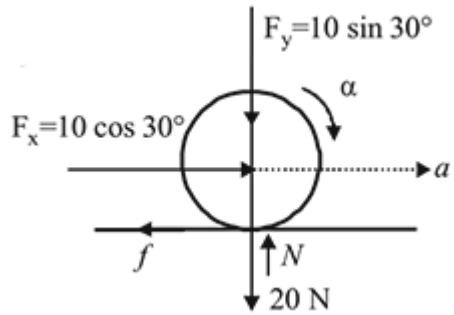
$\tau = I\alpha$

$\Rightarrow fr = \frac{2}{3} \times 2 \times r^2 \times \alpha \ldots$ **(ii)**

(where r is radius of sphere) From (i) and (ii), we get $f = 2\sqrt{3}$ newton,

$N = 20 + 10\sin 30^\circ = 25$

$f = \mu N \Rightarrow \mu = \frac{f}{N} = \frac{2\sqrt{3}}{25} = 0.08 \times \sqrt{3}$



Q.16 Answer:

$\frac{\pi}{15}$Nm

Solution:

$$\tau \times \Delta t = L_0 \qquad \{\because \text{ since } L_f = 0\}$$

or $\tau \times 30 = 2\pi$

$$\tau = \frac{\pi}{15}\ \text{N} - \text{m}$$

Q.17 Answer:

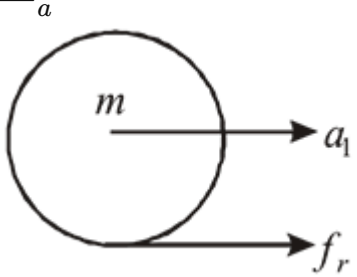
μg

Solution:

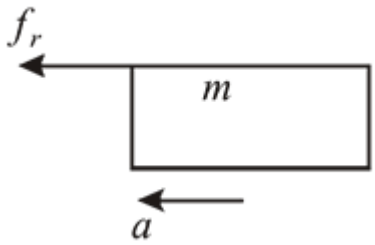
FBD for sphere and block $a_1 = \frac{f_r}{m} = \frac{\mu mg}{m}, a_2 = \frac{f_r}{m} = \frac{\mu mg}{m}$

$\vec{a}_1 = \mu g\hat{i}, \vec{a}_2 = -\mu g\hat{i}$

$\xleftarrow{f_r}_{a}$



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Q.18 Answer:

they will never meet

Solution:

Case 1: When they rotate in same sense $2m\pi = 2\omega t$

$\frac{3\pi}{2} + 2n\pi = \omega t; 2m\pi = 2\left(\frac{3\pi}{2} + 2n\pi\right)$

$2m = 3 + 4n; \quad m = \frac{3}{2} + 2n \Rightarrow m - 2n = \frac{3}{2}$

Not possible for m **and** n **being integer. Case** 2 **: When they rotate in opposite sense** $2m\pi = 2\omega t$

$\frac{\pi}{2} + 2n\pi = \omega t; 2m\pi = 2\left(\frac{\pi}{2} + 2n\pi\right)$

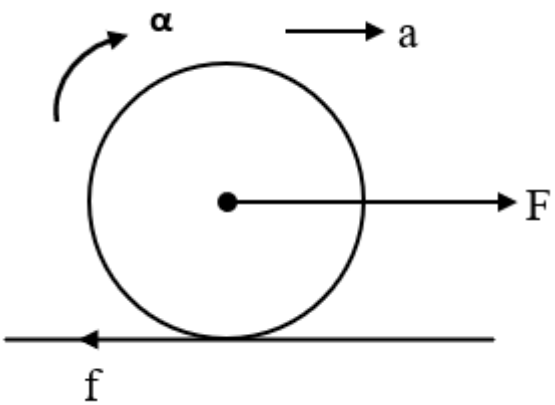
$2m\pi = \pi + 4n\pi; 2m - 4n = 1$

Not possible for m **and** n **integer.**

Q.19 Answer:

none of these

Solution:



Assuming the surface rough enough, so that pure rolling occurs.

Let 'α**' be the angular acceleration**

we get, $F - f = Ma$ **(1)**

and $f \times R = \frac{2}{5}MR^2$ **(2) Also,** $a = \alpha R$ **(3)**

Using (1), (2) and (3)

$F = \frac{7}{5}\mathrm{Ma} \quad \text{or}, \quad a = \frac{5\,\mathrm{F}}{7\,\mathrm{m}}$

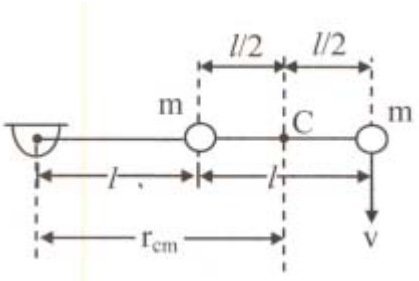
Also $= f = \frac{2}{7}\,\mathrm{F}$**, Hence Option D is correct**

Q.20 Answer:

9

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Solution:



Distance of centre of mass from pivot, $r_{cm} = \frac{m\,l + m(2l)}{m+m} = \frac{3}{2}\,l$

Moment of inertia of system about centre of mass, $I_{cm} = m\left(\frac{l}{2}\right)^2 + m\left(\frac{l}{2}\right)^2 = \frac{m\,l^2}{2}$

Angular velocity of rod , $\omega = \frac{v}{2l}$

Velocity of centre of mass,

$v_{cm} = \omega\frac{3}{2}\,l = \frac{v}{2l}\frac{3}{2}\,l = \frac{3}{4}\,v$

Translational kinetic energy of system,

$K_{translational} = \frac{1}{2}\,M_{treal}\,v_{cin}^2$

$= \frac{1}{2}(2m)\frac{3}{4}\,v^2 = \frac{9}{16}\,mv^2$

Rotational kinetic energy of system,

$K_{rotational} = \frac{1}{2}\,I_{cm}\,\omega^2$

$= \frac{1}{2}\,\frac{m\,l^2}{2}\,\frac{v}{2l}^2 = \frac{1}{16}\,mv^2$

$\therefore \quad \frac{K_1}{K_r} = \frac{\frac{9mv^2}{16}}{\frac{mv^2}{16}} = 9$

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